

LLM-Guided Dynamic Adaptation for Temporal Knowledge Graph Reasoning

LLM-DA (NeurIPS 2024) introduces a hybrid symbolic–language model approach for reasoning over **Temporal Knowledge Graphs (TKGs)** — graphs where facts (edges) evolve over time. The framework combines **rule-based inference** with **LLM-guided rule adaptation**, enabling models to update their reasoning patterns dynamically without full retraining.

Temporal Knowledge Graphs

A TKG represents facts as time-stamped quadruples:

(h, r, t, τ)

Example:

`(Obama, president_of, USA, 2008)`

As time advances, the graph evolves with new entities, relationships, and patterns — forming a **dynamic graph**.

Overview of the LLM-DA Framework

LLM-DA is a **3-step system**:

1. Rule Extraction

- Use a symbolic rule miner (e.g., AnyBURL) to extract logical rules from the TKG at time t_t

Rules are in Horn form:

$\text{works_at}(x, y) \wedge \text{located_in}(y, z) \rightarrow \text{lives_in}(x, z)$

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- These rules capture compositional reasoning patterns from historical data

2. LLM-Based Rule Generation

- Translate symbolic rules into natural language

“If a person works at a company, and the company is in a city, they likely live in that city.”

- Provide the LLM with context about recent graph changes (e.g., rise in remote work)
- Ask the LLM to revise the rule based on new trends

“Only assume this if remote work is not mentioned.”

Translate the revised rule back to symbolic form:

$\text{works_at}(x, y) \wedge \text{located_in}(y, z) \wedge \neg \text{remote}(x, y) \rightarrow \text{lives_in}(x, z)$

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This allows for **rule adaptation without retraining**, guided by the LLM’s understanding of evolving semantics.

3. Rule Adaptation and Deployment

- Apply adapted rules to the updated TKG at time $t+1$
- Use them to:
 - Predict missing or future links
 - Detect which rules have become inaccurate
 - Induce new, temporally aligned rules

This creates a **feedback loop** where symbolic reasoning evolves with the data stream.

Results

- Evaluated on ICEWS14, ICEWS05-15, and YAGO datasets
 - LLM-DA significantly outperforms static rule-based baselines (e.g., Ruleformer) and neural models (e.g., TComplEx, TNTComplEx) under temporal distribution shift
 - Maintains high accuracy while reducing the need for retraining or re-mining from scratch
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Why It Matters for the Survey

LLM-DA is one of the first frameworks to:

- Use LLMs to **edit and refine symbolic rules** for dynamic graph reasoning
- Enable **time-sensitive adaptation** of inference strategies
- Bridge **language understanding and logical inference** in evolving TKGs

This paper adds an important dimension to LLM×dynamic graph research by showing how **LLMs can guide symbolic reasoning to remain robust over time.**